

SUYOG KARKI

Data Science Intern Candidate | ML & Analytics

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PROFILE

Data Science student experienced in building end-to-end machine learning projects using Python, SQL, and Scikit-learn — covering data cleaning, EDA, feature engineering, preprocessing pipeline, cross-validation, hyperparameter tuning, and model evaluation. Applied this across projects in weather forecasting, customer churn, and healthcare risk. I'm genuinely curious about AI and machine learning, and I want to keep getting better at it — looking for a data science/ML or analytics internship to learn and grow.

EDUCATION

Higher Secondary Education (+2) - Science, 2020 – 2022
Kathmandu Model College (KMC), Bagbazar
GPA: 3.35 / 4.0

Bachelor of Science in Information Technology (BSIT) - Data Science, 2023 – 2027
Kings College, Kathmandu (Affiliated with Westcliff University, California)
Currently in 4th Year
GPA: 3.54 / 4.0

TECHNICAL SKILLS

- Languages : Python · SQL
- Data Analysis : Pandas · NumPy · Excel
- Data Visualization: Matplotlib · Seaborn · Power BI
- Tools: Git & GitHub
- Machine Learning : Scikit-learn

CERTIFICATIONS

- Supervised Learning with Scikit-learn | DataCamp | 2026
- Associate Data Scientist Certification | DataCamp | 2026
- SQL Associate | DataCamp | 2026

PROJECTS

Chini — Diabetes Risk Predictor for Women

Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn · SciPy · Streamlit · Docker

GitHub: https://github.com/suyogkarki1/diabetes_pipeline.git [↗](#)

Live: <https://chinisugar.streamlit.app/> [↗](#)

- Built a diabetes risk predictor for adult women on the **Pima Indians dataset** (768 records), using a scikit-learn pipeline that fills in missing values and scales features, so it works even with incomplete patient data.
- Used statistical tests (**Welch's t-test**, **Cohen's d**) to find which measurements best separate diabetic from non-diabetic patients.

- Compared five models with 5-fold cross-validation and tuned each with GridSearchCV (including L1/L2/ElasticNet regularization for Logistic Regression), keeping all preprocessing inside the folds to prevent data leakage.
- Chose **Logistic Regression** and scored on PR-AUC instead of accuracy because the data is imbalanced, reaching **0.81** ROC-AUC, **73%** accuracy, and **0.72** recall on the test set.
- Deployed it as an interactive Streamlit web app, containerized with Docker for easy setup.

Kathmandu Rain Prediction

Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · Streamlit

GitHub: <https://github.com/suyogkarki1/Rain-Prediction-> 

Live: <https://barshamausam.streamlit.app/> 

- Sourced a 10-year Kathmandu weather dataset (2015–2025, 3,468 records) from Kaggle, and performed data cleaning, unit conversion, and feature engineering across 14 meteorological variables for supervised classification.
- Built and evaluated four classification models — Logistic Regression, Decision Tree, K-Nearest Neighbors, and Random Forest — using stratified 6-fold cross-validation; selected Random Forest as the final model, achieving **84% accuracy and 0.90 ROC-AUC** on held-out test data.
- Conducted feature importance analysis identifying **humidity, temperature, and cloud cover** as the strongest rainfall predictors, validating the model against real-world Kathmandu Valley monsoon patterns.
- Applied **bootstrap resampling** to estimate 2026 monthly rain probabilities with lower and upper uncertainty bounds, and developed a separate **SARIMA** seasonal time-series model to forecast monthly rainy-day percentages for comparison.
- Deployed the trained model as an interactive Streamlit web application (**SkyCast**) enabling real-time rain classification from user-defined weather inputs and visualizing seasonal forecast trends.

Customer Churn Prediction

Python · Pandas · Scikit-learn · Matplotlib · Seaborn

GitHub: <https://github.com/suyogkarki1/customer-churn-prediction> 

- Analyzed a Kaggle dataset of 10,000 European bank customers to identify behavioral patterns linked to churn, including credit score, account balance, and product usage habits.
- Engineered new features including **BalancePerProduct** and **AgeGroup**; conducted full EDA to uncover class imbalance and key churn indicators before applying preprocessing steps.
- Achieved **86.35% accuracy and 0.87 ROC-AUC** using Gradient Boosting Classifier, selected after benchmarking against simpler models including Logistic Regression and Decision Tree.
- Feature importance analysis revealed **Age and NumOfProducts** as the strongest churn predictors — suggesting retention teams should prioritize older single-product customers as the highest risk segment

Smart Healthcare and Lifestyle Prediction

Python · Pandas · Scikit-learn · Matplotlib · Seaborn

GitHub: <https://github.com/suyogkarki1/Smart-Healthcare-and-Lifestyle-Prediction> 

- Worked with a Kaggle healthcare dataset of 5,000 patient records containing clinical indicators, including blood pressure, cholesterol, BMI, smoking habits, and chest pain severity.
- Performed end-to-end pipeline — data cleaning, EDA, feature analysis, model training, and evaluation — comparing multiple classifiers before finalizing Logistic Regression.

- Achieved **99.4% accuracy and 0.999 ROC-AUC**; identified that well-separated class distributions in the dataset contributed to high performance and validated results using cross-validation to rule out data leakage.
- Feature importance analysis revealed **Age, BMI, and Smoking** as the strongest heart disease predictors — insights that can help clinicians prioritize high-risk patients for early screening and preventive intervention.

UAE ID Card Number Extraction Pipeline

Python · YOLOv11 (Ultralytics) · OpenCV · Tesseract OCR · PostgreSQL

Github: https://github.com/suyogkarki1/object_detection.git 

- Annotated a custom dataset of 424 UAE ID card images on Ultralytics HUB, labeling the ID-number field as a dedicated detection class, split into 405 train / 10 validation / 9 test images.
- Trained and deployed a custom YOLOv8 model via Ultralytics' hosted endpoint, achieving **99.5% mAP50** and **95.0% recall** on ID-number localization.
- Built an OCR pipeline (OpenCV preprocessing + Tesseract) to extract and clean detected ID numbers via regex, then persisted records to PostgreSQL with conflict-safe deduplication.

LANGUAGES

Nepali

Native

English

Fluent